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科目: 概率論與隨機過程

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$$1. (a) p_N(n) = \frac{\lambda^n}{n!} e^{-\lambda}, \quad f_{X_i}(x) = \begin{cases} 1, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

$$E[e^{sY} | N=n] = \prod_{i=1}^n E[e^{sX_i}] = (E[e^{sX}])^n$$

$$E[e^{sX}] = \int_0^1 e^{sx} dx = \frac{1}{s}(e^s - 1)$$

$$M_Y(s) = E[e^{sY}] = E_N[E[e^{sY} | N]] = E\left[\left(\frac{e^s - 1}{s}\right)^N\right]$$

$$= \sum_{n=0}^{\infty} \left(\frac{e^s - 1}{s}\right)^n \cdot \frac{\lambda^n}{n!} e^{-\lambda} = e^{\frac{\lambda}{s}(e^s - 1)} \cdot e^{-\lambda} = e^{\lambda\left(\frac{e^s - 1}{s} - 1\right)}$$

$$(b) E[Y] = \frac{dM_Y(s)}{ds} \Big|_{s=0} = \lim_{s \rightarrow 0} \left[e^{\lambda\left(\frac{e^s - 1}{s} - 1\right)} \cdot \lambda \cdot \left(\frac{e^s - 1}{s^2}\right) \right]$$

$$= \lambda \lim_{s \rightarrow 0} \left[e^{\lambda\left(\frac{1+s-1}{s} - 1\right)} \cdot \frac{e^s s + e^s - e^s}{2s^2} \right]$$

$$= \frac{\lambda}{2}$$

$$(c) E[Y] = E[E[Y | N]] = E[N E[X]] = E[N] E[X]$$

$$E[N] = \sum_{n=0}^{\infty} n \cdot \frac{\lambda^n}{n!} e^{-\lambda} = \sum_{n=1}^{\infty} \frac{\lambda^n}{(n-1)!} e^{-\lambda} = \lambda$$

$$E[X] = \frac{1}{2}$$

$$\Rightarrow E[Y] = \lambda \cdot \frac{1}{2} = \frac{\lambda}{2}, \text{ 與 (b) 的結果}$$

2. (a) Let the waiting time at i th light is X_i ($i=1, 2, 3, 4$)

$$X = X_1 + X_2 + X_3 + X_4, \quad p_{X_i}(0) = \frac{1}{2}, \quad f_{X_i}(x) = \frac{1}{\frac{1}{2}\sqrt{2\pi}} \exp\left\{-\frac{(x+1)^2}{2x(\frac{1}{2})^2}\right\}, \quad x > 0$$

$$f_{X_i}(x) = \begin{cases} \frac{1}{\frac{1}{2}\sqrt{2\pi}} \exp\left\{-\frac{(x+1)^2}{2x(\frac{1}{2})^2}\right\}, & x > 0 \\ 0, & x < 0 \end{cases}$$

$$M_{X_i}(s) = \exp\left\{\frac{0 \cdot s^2}{2} + \mu_i s\right\}$$

$$M_X(s) = M_{X_1}(s) M_{X_2}(s) M_{X_3}(s) M_{X_4}(s)$$

$$= \exp\left\{\frac{40s^2}{2} + 4\mu_i s\right\}$$

$$= \exp\left\{\frac{s^2}{2} + 4s\right\}$$

$$\Rightarrow f_X(x) = \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{(x-4)^2}{2}\right\}$$

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2(a) Let X_i be the waiting time at i th light, K be the number of red lights

$$X = X_1 + X_2 + X_3 + X_4$$

$$f_{X|K}(x|k) = N(x, \frac{k}{4})$$

$$f_X(x) = \sum_{k=1}^4 \left(\frac{1}{4}\right) \left(\frac{1}{2}\right)^4 N(x, \frac{k}{4}), \quad M_X(s) = \sum_{k=1}^4 \left(\frac{1}{4}\right) \left(\frac{1}{2}\right)^4 \exp\left\{-\frac{1}{2} \cdot \frac{1}{k} \cdot s^2 + ks\right\}$$

X isn't normal, as the mixture of normal distribution with different μ and σ is not a normal RV.

Meanwhile, we can only find the table or use computer to compute $P(X > 4)$

$$(b) M_X(s) = E(e^{s(X_1+X_2+X_3+X_4)}) = (M_{X_1}(s))^4$$

$$M_{X_1}(s) = E[e^{sX_1}] = \frac{1}{2} \cdot e^{s \cdot 0} + \frac{1}{2} \cdot \int_{-\infty}^{+\infty} \frac{1}{\sigma_1 \sqrt{2\pi}} \exp\left\{-\frac{(x-\mu_1)^2}{2\sigma_1^2}\right\} e^{sx} dx$$

$$= \frac{1}{2} + \frac{1}{2} \exp\left\{\frac{\sigma_1^2 s^2}{2} + \mu_1 s\right\} = \frac{1}{2} + \frac{1}{2} \exp\left\{\frac{1}{8} s^2 + s\right\}$$

$$\Rightarrow M_X(s) = \left[\frac{1}{2} + \frac{1}{2} \exp\left(\frac{1}{8} s^2 + s\right)\right]^4$$

$$3. (a) E[W] = E[E[W|M]] = E\left[\sum_{k=1}^M E[K_k|M]\right] = E[M E[K]] = E[M] E[K]$$

$$\begin{aligned} \text{var}(W) &= E[\text{var}(W|M)] + \text{var}[E[W|M]] \\ &= E[M \text{var}(K)] + \text{var}[E[K] M] \\ &= E[M] \text{var}(K) + (E[K])^2 \text{var}(M) \end{aligned}$$

$$(b) E[Y] = E[E[Y|W]] = E[W E[X]] = E[W] E[X] = E[M] E[K] E[X]$$

$$\begin{aligned} \text{var}(Y) &= E[W] \text{var}(X) + (E[X])^2 \text{var}(W) \\ &= E[M] E[K] \text{var}(X) + (E[X])^2 [E[M] \text{var}(K) + (E[K])^2 \text{var}(M)] \end{aligned}$$

(c) Let Y be the total weight of crate, X_i be the i th carton, W_{ij} be the weight of the j th widget in the i th carton.

$$Y = X_1 + X_2 + \dots + X_M, \quad X_i = W_{i1} + W_{i2} + \dots + W_{ik_i}$$

$$E[Y] = E[E[Y|M]] = E[M] E[X] = E[M] E[E[X|K]] = E[M] E[K] E[W]$$

$$= \frac{1}{p} \cdot \mu \cdot \frac{1}{\lambda} = \frac{\mu}{p\lambda}$$

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$$\begin{aligned}
 \text{var}(Y) &= E[M] \text{var}(X) + (E[X])^2 \text{var}(M) \\
 &= E[M] (E[K] \text{var}(W) + (E[W])^2 \text{var}(K)) + [E[K] E[W]]^2 \text{var}(M) \\
 &= \frac{1}{p} \left(\mu \cdot \frac{1}{\lambda^2} + \frac{1}{\lambda^2} \cdot \mu \right) + \left[\mu \cdot \frac{1}{\lambda} \right]^2 \cdot \frac{1-p}{p^2} \\
 &= \frac{2\mu}{p\lambda^2} + \frac{\mu(1-p)}{\lambda p^2}
 \end{aligned}$$

4. (a) Let $L = X+Y$, $R = X+Y+Z = S+Z$

~~$$f_L(l) = \int_0^{l-1} f_X(x) f_Y(l-x) dx = \int_0^l f_Y(l-x) dx$$~~

4. (a) Let $L = X+Y$, $R = L+Z$

$$f_L(l) = \int_0^1 f_X(x) f_Y(l-x) dx = \int_0^1 f_Y(l-x) dx$$

When $l \geq 2$, $l-x \geq 1 \Rightarrow f_Y(l-x) = 0 \Rightarrow f_L(l) = 0$

When $1 \leq l < 2$, $f_L(l) = \int_{l-1}^1 f_Y(l-x) dx = 2-l$

When $0 \leq l < 1$, $f_L(l) = \int_0^l f_Y(l-x) dx = l$

When $l < 0$, $f_L(l) = 0$

$$f_R(r) = \int_0^1 f_Z(z) f_L(r-z) dz = \int_0^1 f_L(r-z) dz$$

When $r \geq 3$, $r-z \geq 2 \Rightarrow f_L(r-z) = 0 \Rightarrow f_R(r) = 0$

When $2 \leq r < 3$, $f_R(r) = \int_{r-2}^1 f_L(r-z) dz = \int_{r-2}^1 (z-r+2) dz = \frac{1}{2} (z-r+2)^2 \Big|_{r-2}^1 = \frac{(3-r)^2}{2}$

When $1 \leq r < 2$, $f_R(r) = \int_0^{r-1} f_L(r-z) dz + \int_{r-1}^1 f_Z(r-z) dz$

$$= \int_0^{r-1} (z+2-r) dz + \int_{r-1}^1 (r-z) dz$$

$$= \frac{1}{2} (z+2-r)^2 \Big|_0^{r-1} + \frac{1}{2} (r-z)^2 \Big|_{r-1}^1$$

$$= \frac{1}{2} [1 - (2-r)^2 + 1 - (r-1)^2] = \frac{-2r^2 + 6r - 3}{2}$$

When $0 \leq r < 1$, $f_R(r) = \int_0^r f_Z(r-z) dz = \int_0^r (r-z) dz = \frac{1}{2} (r-z)^2 \Big|_r^0 = \frac{r^2}{2}$

When $r < 0$, $f_R(r) = 0$

(b) When $n=1$, $f_{x_1}(x)=1$, $\deg(f_{x_1}(x))=0=n-1$, statement corrects

Assume that ~~the~~ the statement still ~~is~~ correct when $n=k \in \mathbb{Z}^+$

Let $S_k = \sum_{i=1}^k x_i$, i.e. $\deg(f_{S_k}(s)) = k-1$

$$f_{S_{k+1}}(s) = \int_0^1 f_{S_k}(s-x) dx$$

When $n=k+1$,

$$f_{S_{k+1}}(s) = \int_0^1 f_{x_{k+1}}(x) f_{S_k}(s-x) dx = \int_0^1 f_{S_k}(s-x) dx$$

Because $\deg(f_{S_k}(s-x)) = \deg(f_{S_k}(x)) = k-1$

$$\Rightarrow \int_0^1 f_{S_k}(s-x) dx \quad \deg \left[\int_0^1 f_{S_k}(s-x) dx \right] = \deg[f_{S_k}(s-x)] + 1 = \deg[f_{S_k}(x)] + 1 \\ = k-1+1 = k$$